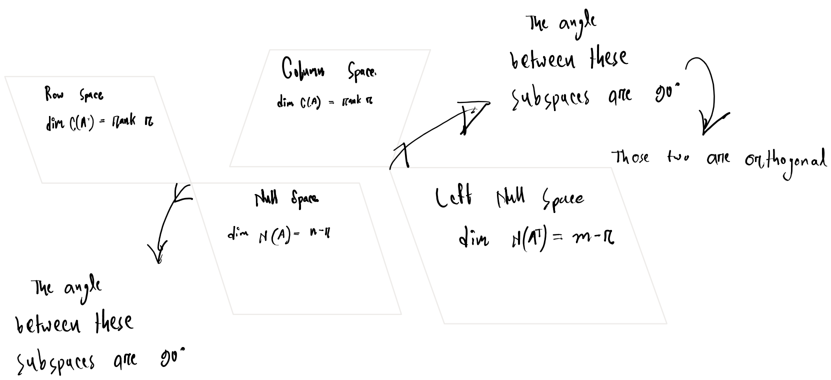
Orthogonal Vectors & Subspaces

Nullspace $\perp$ Rowspace

$N(A^TA) = N(A)$



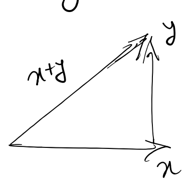
↳ These two are orthogonal

But what does it mean for two vectors to be orthogonal?

Orthogonal Vectors

↳ Orthogonal is just another word for perpendicular

↳ In n dimensional space orthogonal vectors are where the angle between them is 90°.

↳ 

(pythagoras)

↳ How do I know these two are orthogonals? → $x^Ty = 0$

$\|x\|^2 + \|y\|^2 = \|x+y\|^2$ → Only true for right angle triangles?

Example: $x = \begin{bmatrix}1\\2\\3\end{bmatrix}$, $y = \begin{bmatrix}2\\-1\\0\end{bmatrix}$, $x+y = \begin{bmatrix}3\\1\\3\end{bmatrix}$

$\|x\|^2 = 14$, $\|y\|^2 = 5$, $\|x+y\|^2 = 19$

Now, from $\|x\|^2 + \|y\|^2 = \|x+y\|^2$

$\Rightarrow x^Tx + y^Ty = (x+y)^T(x+y)$

$\Rightarrow x^Tx + y^Ty = x^Tx + x^Ty + xy^T + yy^T$

$\Rightarrow x^T y + x y^T = 0$ $\quad [x^T y = x y^T]$

↳ Dot product is symmetric.

$\Rightarrow 2x^T y = 0$

$\Rightarrow x^T y = 0$

Assuming, Subspace S is orthogonal to subspace T

↳ it means every vector in S is orthogonal to every vector in T.

* Imagine a plane, what are their possible subspaces? Zero vector, a line through the origin and the whole plane.

↳ Now, when is a line through the origin is orthogonal to the plane? $\boxed{\text{Never}}$

↳ Now, when is a line through the origin is orthogonal to the zero subspace? $\boxed{\text{Always}}$

↳ Now, when is a line through the origin is orthogonal to a different line through the origin? ⇝ $\boxed{\text{Two lines have to meet at } 90^\circ.}$

* We know that row space is orthogonal to nullspace

But why?

$Ax = 0$. Here x is null space of A.

$$\begin{bmatrix} \text{row1 of } A \\ \text{row2 of } A \\ \vdots \\ \text{row } m \text{ of } A \end{bmatrix} \begin{bmatrix} x \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

↳ $(\text{row1 of } A) \cdot x = 0$

$(\text{row2 of } A) \cdot x = 0$

$\vdots$

$(\text{row } m \text{ of } A) \cdot x = 0$

$\Rightarrow$ So, x is orthogonal to all the rows.

Now,

$c_1 (\text{row1})^T x = 0$

$c_2 (\text{row2})^T x = 0$

↳ $(c_1 \text{row}_1 + c_2 \text{row}_2)^T x = 0$

* Nullspace and rowspace are orthogonal complements in $\mathbb{R}^n$.

Nullspace contains all vectors that are perpendicular to row space.

Now, if a system of equations are $Ax = b$.

$(m > n)$

Now, if a system of equations are $Ax=b$.
when there is no solution, then "Solve" it. $m\times n$ $(m>n)$

Now, let's take a look at A^TA.

- if, $A \to (m\times n)$ then A^TA is $n\times n$
- Also, A^TA is symmetric. How? $(A^TA)^T = A^T \cdot (A^T)^T = A^TA$.

Now from that, $Ax=b$

$\Rightarrow A^TA\hat{x} = A^Tb$

let's take an example: $Ax = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

$\to$ this can only be solved if b is a linear combination of A but usually it won't be.

So, $A^TA = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 3 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 1 & 3 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 9 \\ 9 & 27 \end{bmatrix}$

$\to$ not invertible.

$\to$ cannot be solved

Some key takeaways:

- $N(A^TA) = N(A)$
- $\text{rank}(A^TA) = \text{rank}(A)$
- So, A^TA is invertible exactly if A has independent columns.
  - $\to$ A square matrix is invertible if and only if its nullspace contains only the zero vector. This means that the only solution to $Ax=0$ is $x=0$.
  - $\to$ If there is a nonzero vector x such that $Ax=0$, then A has dependent columns and cannot have an inverse.

Fundamental Theorem of Linear Algebra

Part 2:

- $N(A)$ is the orthogonal complements of the row space $C(A^T)$ in R^n.

- $N(A^T)$ is the orthogonal complement of the column space $C(A)$ in R^m.

- If V and W are orthogonal subspaces in R^n then $\dim(V) + \dim(W) \le n$.
  - $\to$ Why?
    - Orthogonal subspaces do not share any nonzero vectors.

↳ Why?

- Orthogonal subspaces do not share any nonzero vectors.
- If their dimensions exceeded n, it would imply that some vectors must be shared, contradicting orthogonality.
- The only vector in two orthogonal subspaces is the $\boxed{\text{zero vector.}}$

⊛ The orthogonal complement of a subspace V, denoted $V^\perp$, consists of all vectors that are perpendicular to every vector in V.

↳ - If V is a plane in $\mathbb{R}^3$, then $V^\perp$ is the line perpendicular to the plane.
- If the total space is $\mathbb{R}^n$, then: $\dim V + \dim V^\perp = n$.
- Every vector in $\mathbb{R}^n$ can be uniquely decomposed as a sum of:
  - A component in V.
  - A component in $V^\perp$.

⊛ For an $n \times m$ matrix A:

- The row space of A, denoted $R(A)$ → $C(A^T)$, consists of all linear combinations of its rows.
  The null space, $N(A)$ consists of all solutions to $Ax=0$.
- And their subspaces are orthogonal complements, meaning any vector in $\mathbb{R}^n$ can be decomposed as: $\boxed{x = x_{row} + x_{null}}$

  ↳ $x = x_r + x_n$

  Their dimensions add up to n: $\underbrace{r}_{\text{rank (dimension of the row space)}} + \underbrace{(n-r)}_{\text{nullity (dimension of the null space)}} = n$

- Same can be applied for column space and left null space.
  The column space of A, $C(A)$, consists of all linear combinations of its columns.
  The left null space of A, $N(A^T)$, consists of all solutions to $A^Ty=0$
- And for their subspaces: $\boxed{y = y_{col} + y_{null}}$ ⇝ $y = y_c + y_n$

$$\underbrace{r}_{\text{rank (dimension of the column space)}} + \underbrace{(m-r)}_{\text{nullity (dimension of the left nullspace)}} = m$$

row × column

$n \times m$

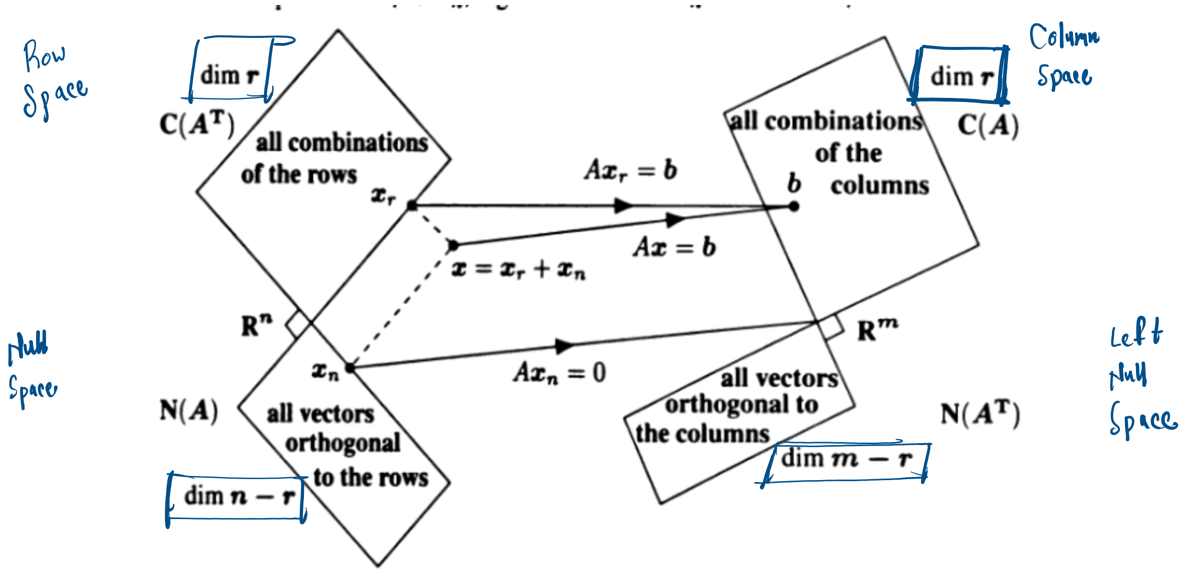


Figure 4.1: $Ax_r = b$ is in the column space and $Ax_n = 0$. The complete solution to $Ax = b$ is $x =$ one $x_r +$ any x_n. Then $||x||^2 = ||x_r||^2 + ||x_n||^2$ and the **minimum norm** solution to $Ax = b$ is $x = x_r$ in the row space plus $x_n = 0$ from $\mathbf{N}(A)$.

→ Smallest possible $\|x\|$ is achieved by taking only the row space component, without adding anything from the null space.

→ If $Ax = b$, and $x = x_r + x_n$.

then $Ax_r = b$, $Ax_n = 0$

✱ Every vector b in the column space comes from exactly one vector x_r in the row space.

→ Proof: If $Ax_r = b$, the b is in column space of A.

If there were two different vectors x_r and x_r' in the row space that give the same result when multiplied by A,

then: $$Ax_r = Ax_r'$$

$\hookrightarrow A(x_r - x_r') = 0$

$\Rightarrow x_r - x_r' = 0$

$\hookrightarrow$ Meaning the difference $x_r - x_r'$ is in the null space of A, because it gets mapped to the zero vector by A.

Therefore, $x_r = x_r'$

(N) Every matrix of rank r has an r by r invertible submatrix.

Proof: Let's say a matrix A has a submatrix with r independent columns. Then, C and C^T have both rank r.

So, C^T has r independent columns - in other words C has r independent rows. This locates an r by r invertible matrix of A.

Combining Bases from Subspaces

$\hookrightarrow$ A basis contains linearly independent vectors that span the space.

$\hookrightarrow$ Any n independent vectors in R^n must span R^n. So they are a basis.
Any n vectors that span R^n must be independent. So they are a basis.

Example 4 **Suppose S is a six-dimensional subspace of nine-dimensional space R^9.**

(a) What are the possible dimensions of subspaces orthogonal to S? **0, 1, 2, 3**

(b) What are the possible dimensions of the orthogonal complement $S^\perp$ of S? **3**

(c) What is the smallest possible size of a matrix A that has row space S? **6 by 9**

(d) What is the smallest possible size of a matrix B that has nullspace $S^\perp$? **6 by 9**

Sol^n:

a) S is a six-dimensinal subspace of nine-dimensional space R^9.

Here, $\dim(S) + \dim(S^\perp) \le 9$

$\Rightarrow \dim(S^\perp) \le 9 - 6$

$\Rightarrow \dim(S^\perp) \le 3$ $\Rightarrow$ So, possible dimension are: 0, 1, 2, 3.

b) For, orthogonal complement, $\dim(S) + \dim(S^\perp) = 9$

$\Rightarrow \dim(S^\perp) = 9 - 6 = 3$.

c) S is a six-dimensinal subspace

$\Rightarrow$ to span S, must have 6 independent rows.

S is a subspace of R^9

$\Rightarrow$ each row vector must have 9 components.
meaning A must have 9 columns (one for each co-ordinate in R^9)

So, the smallest possible size of a matrix A is 6×9.

(from b)

So, the smallest possible ...

d) $S^{\perp}$ is a three dimensional subspace (from b)

↳ So null space's dimension of B is 3 $(n-r)$

The matrix B is in $\mathbb{R}^9$

↳ So B must have 9 columns.

From rank-nullity theorem, $r + (n-r) = 9$

$\hookrightarrow r + 3 = 9$;

$\hookrightarrow r = 6$. ⇝ six independent rows.

So, smallest possible matrix B is 6×9.

Example Problem: A subspace S is spanned by $(1\ 2\ 2\ 3)$ and $(1\ 3\ 3\ 2)$

(i) Find a basis for $S^{\perp}$

(ii) Can every v in $\mathbb{R}^4$ be written uniquely in terms of S and $S^{\perp}$?

Sol^n: (i) If x is $S^{\perp}$, then x is orthogonal to every vector in S.

So, $(1\ 2\ 2\ 3)x = 0$

$(1\ 3\ 3\ 2)x = 0$

$$\hookrightarrow \begin{bmatrix} 1 & 2 & 2 & 3 \\ 1 & 3 & 3 & 2 \end{bmatrix} x = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 2 & 3 \\ 1 & 3 & 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

Now, $A = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 1 & 3 & 3 & 2 \end{bmatrix}$

$$R(A) = \begin{array}{cccc} & & & \\ \end{array}\begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

pivot pivot free free

$$\Rightarrow x_1 + 2x_2 + 2x_3 + 3x_4 = 0 \quad \text{--- (i)}$$

$$x_2 + x_3 - x_4 = 0 \quad \text{--- (ii)}$$

Assuming $x_3 = 1$ and $x_4 = 0$.

From (ii) $\Rightarrow$ $x_2 = -1$

From (i) $\Rightarrow$ $x_1 = +2 - 2 = 0$

Assuming, $x_3 = 0$ and $x_4 = 1$.

From (ii) $\Rightarrow$ $x_2 = 1$

From (i) $\Rightarrow$ $x_1 = -2 - 3 = -5$

$$\therefore \quad x = C_1 \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} + C_2 \begin{bmatrix} -5 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$\Rightarrow$ basis for $S^{\perp}$.

(ii) Yes.

a vector $v = \underbrace{C_1 \begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \end{bmatrix} + C_2 \begin{bmatrix} 1 \\ 3 \\ 3 \\ 2 \end{bmatrix}}_{S} + \underbrace{C_3 \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} + C_4 \begin{bmatrix} -5 \\ 1 \\ 0 \\ 1 \end{bmatrix}}_{S^{\perp}}$

$$\Rightarrow \underset{1\times 4}{V} = \underset{4\times 4}{\begin{bmatrix} 1 & 1 & 0 & -5 \\ 2 & 3 & -1 & 1 \\ 2 & 3 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{bmatrix}} \underset{4\times 1}{\begin{bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{bmatrix}}$$

$\Rightarrow$ linearly independent.

$\Rightarrow$ invertible matrix

$\Rightarrow$ Inverting the matrix will give us unique co-efficients.

$\Rightarrow$ Gives us unique decomposition